

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

$j(x) = mx + 144$

For the linear function j , m is a constant and $j(12) = 18$. What is the value of $j(10)$?

Correct Answer: 39

Rationale

The correct answer is **39**. It’s given that for the linear function j , m is a constant and $j(12) = 18$. Substituting **12** for x and **18** for $j(x)$ in the given equation yields $18 = m(12) + 144$. Subtracting **144** from both sides of this equation yields $-126 = m(12)$. Dividing both sides of this equation by **12** yields $-10.5 = m$. Substituting -10.5 for m in the given equation, $j(x) = mx + 144$, yields $j(x) = -10.5x + 144$. Substituting **10** for x in this equation yields $j(10) = (-10.5)(10) + 144$, or $j(10) = 39$. Therefore, the value of $j(10)$ is **39**.